EXERCISE 8.2

(1) Verify whether the following ratios are direction cosines of some vector or not.

(i) $\frac{1}{5}, \frac{3}{5}, \frac{4}{5}$ (ii) $\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{2}$ (iii) $\frac{4}{3}, 0, \frac{3}{4}$

(2) Find the direction cosines of a vector whose direction ratios are

(i) 1 , 2 , 3 (ii) 3 , − 1 , 3 (iii) 0 , 0 , 7

(3) Find the direction cosines and direction ratios for the following vectors.

(i) $3\hat{i} - 4\hat{j} + 8\hat{k}$ (ii) $3\hat{i} + \hat{j} + \hat{k}$ (iii) $\hat{j}$

(iv) $5\hat{i} - 3\hat{j} - 48\hat{k}$ (v) $3\hat{i} - 3\hat{k} + 4\hat{j}$ (vi) $\hat{i} - \hat{k}$

(4) A triangle is formed by joining the points (1, 0, 0), (0, 1, 0) and (0, 0, 1). Find the direction cosines of the medians.

(5) If $\frac{1}{2}, \frac{1}{\sqrt{2}}, a$ are the direction cosines of some vector, then find a.

(6) If (a , $a + b$, $a + b + c$) is one set of direction ratios of the line joining (1, 0, 0) and (0, 1, 0), then find a set of values of a, b, c.

(7) Show that the vectors

$2\hat{i} - \hat{j} + \hat{k},\ 3\hat{i} - 4\hat{j} - 4\hat{k},\ \hat{i} - 3\hat{j} - 5\hat{k}$ form a right angled triangle.

(8) Find the value of λ for which the vectors $\vec{a} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\vec{b} = \hat{i} + \lambda\hat{j} + 3\hat{k}$ are parallel.

(9) Show that the following vectors are coplanar

(i) $\hat{i} - 2\hat{j} + 3\hat{k},\ -2\hat{i} + 3\hat{j} - 4\hat{k},\ -\hat{j} + 2\hat{k}$

(ii) $2\hat{i} + 3\hat{j} + \hat{k},\ \hat{i} - \hat{j},\ 7\hat{i} + 3\hat{j} + 2\hat{k}$.

(10) Show that the points whose position vectors $4\hat{i} + 5\hat{j} + \hat{k},\ -\hat{j} - \hat{k},\ 3\hat{i} + 9\hat{j} + 4\hat{k}$ and $-4\hat{i} + 4\hat{j} + 4\hat{k}$ are coplanar.

(11) If $\vec{a} = 2\hat{i} + 3\hat{j} - 4\hat{k},\ \vec{b} = 3\hat{i} - 4\hat{j} - 5\hat{k}$, and $\vec{c} = -3\hat{i} + 2\hat{j} + 3\hat{k}$, find the magnitude and direction cosines of (i) $\vec{a} + \vec{b} + \vec{c}$ (ii) $3\vec{a} - 2\vec{b} + 5\vec{c}$.

(12) The position vectors of the vertices of a triangle are $\hat{i} + 2\hat{j} + 3\hat{k};\ 3\hat{i} - 4\hat{j} + 5\hat{k}$ and $-2\hat{i} + 3\hat{j} - 7\hat{k}$. Find the perimeter of the triangle.

(13) Find the unit vector parallel to

$3\vec{a} - 2\vec{b} + 4\vec{c}$ if $\vec{a} = 3\hat{i} - \hat{j} - 4\hat{k},\ \vec{b} = -2\hat{i} + 4\hat{j} - 3\hat{k}$, and $\vec{c} = \hat{i} + 2\hat{j} - \hat{k}$.

(14) The position vectors $\vec{a}, \vec{b}, \vec{c}$ of three points satisfy the relation $2\vec{a} - 7\vec{b} + 5\vec{c} = \vec{0}$. Are these points collinear?

(15) The position vectors of the points P, Q, R, S are $i + j + k,\ 2\hat{i} + 5\hat{j},\ 3\hat{i} + 2\hat{j} - 3\hat{k}$, and $\hat{i} - 6\hat{j} - \hat{k}$ respectively. Prove that the line PQ and RS are parallel.

(16) Find the value or values of m for which $m\ (\hat{i} + \hat{j} + \hat{k})$ is a unit vector.

(17) Show that the points A (1, 1, 1), B(1, 2, 3) and C(2, − 1, 1) are vertices of an isosceles triangle.

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